Special Note

Do not submit your work until you have been put into a project group on Canvas. Submitting early will mean that your group members will not be able to see the assignment grade or the grading on the rubric.

How can you tell if you have been put into a project group? See the FAQ section in milestone 0 for more information.

General

As you start to work together, it is worth considering what you want your working relationship to look like. "Who is on a team matters less than how the team members interact, structure their work, and view their contributions". Please think about how you can work together to provide each other to provide a healthy and productive environment. If you feel that problems have come up within your group, please talk to your project TA or instructor as soon as you can.

If the group runs into problems with collaboration, it's very important to identify and address that as soon as possible. To do this, it is important to check-in with your team regularly, and plan to make regular progress throughout the project time so that you can identify any issues early. Document the division of work by listing things such as <person> will do <task> by <time> and send it as an email to everyone or put the document in a place that everyone can access. Make sure to work and check-in in a way that guarantees you will be able to identify problems substantially before the end of the project.

By default (and in particular, if we don't hear about problems until close to the deadline) all team members will receive the same grade for each project.

If you require instructor/TA intervention: contact us 12 days before the deadline, and arrange a meeting 9 days before the deadline with all team members. Upon first notification about group issues, we will meet with the group to discuss the problem and identify what everyone needs to succeed. If things do not improve after one to two weeks, we will take further action which can include, but is not limited to, grade adjustments or group rearrangement.

Pre-Grading

Pre-grading work is the action of looking over an artifact and commenting on the places it is incorrect or not adhering to the posted grading rubric. The teaching team will not pre-grade your work.

While we are always happy to answer specific questions, we do not have the resources to grade a submission repeatedly. Please refer to the posted rubric on Canvas as that is what we will be using to grade your work.

Here are some examples of specific questions:

- We are debating between these two ER diagram designs and here are the pros and cons of each. What would you recommend that we think about when choosing between the designs?
- Our group wants to base our project on <idea> but <this> aspect of the project seems very close to <blacklist topic>. Here are the ways that this topic differs: <list of reasons>. Is this sufficiently different?

Project Topic

You are allowed to choose your own topic as long as it is not something from our lectures or tutorials, another course, the Internet, previous students' work, your co-op term, etc. The work you submit must be something original created for this class during the current semester.

Blacklist – Some project topics that you may not use are:

- Employees and departments (projects, managers, employees) –textbook example
- A bookstore or library (used in previous semesters)
- Car rental service (used in some previous semesters)
- MP3 storage (used in some previous semesters)
- A school setting (students, instructors, departments, courses, etc.) – We're going to use this as a case study in various parts of the course.
- Banking (accounts, customers) – another common example
- Airlines (flights, planes, passengers, destinations) – another common example used by many textbooks
- Retail sales (customers, products, orders, line orders) – another common example.
- Hospital or medical clinic – again, common
- User management (e.g., gym membership management/tracking activities in a gym) – again, common
- Sports league – again, common
- Job board – again, common
- Event management – again, common
- Any other examples presented in the course

- Any project that you have previously done (i.e., you must do a new project and cannot reuse an existing project idea)

Before you decide on a topic, you will want to carefully consider whether your project would be considered as a variation of something on the blacklist. For example, AirBnB is a topic not on the blacklist but the general structure of this is very similar to a car rental (which is on the blacklist). Uber Eats is not on the blacklist but is essentially a retail sales example. We don't want to see a re-skinned version of a blacklist project. If you feel that your project may be considered similar, talk with your project TA by carefully explaining how your project idea will be significantly different from a banned topic.

Project topics that contain elements of multiple blacklist topics stitched together are not allowed. In the event that we deem your project to be too similar to a blacklisted topic, we reserve the right to apply a penalty of our choosing. The penalty will depend on various factors such as the degree of similarity. We will also ask you to redo milestone 1 for no additional credit; you will only be allowed to proceed with the project once we deem your milestone 1 to be suitable and acceptable. It is likely that the turnaround for this will be short as you will need your milestone 1 approved before proceeding to subsequent milestones. There will not be any additional concessions granted for subsequent milestones.

If you are having trouble coming up with a topic, try thinking about your hobbies and interests. Please do not send the teaching team a list of topics to ask them to pick one for you. It is the group's responsibility to choose a suitable topic.

The blacklist is not asking you to invent something completely novel or startup-worthy. Rather, it's designed to give you a better learning experience. Here's why:

Common topics make it harder to learn. When working with frequently-used domains (like library management or e-commerce sites), it's tempting to look at existing solutions and replicate familiar patterns. This can prevent you from engaging deeply with the design challenges that this course is meant to teach. We want you to grapple with real tradeoffs—not just implement a template you've seen before.

There's no such thing as a perfect database design. Every design involves compromises. Milestone 1 is specifically about learning to identify these tradeoffs and make intentional decisions about what to prioritize. This skill is much harder to develop when you're working with a well-worn problem that already has established "standard" solutions.

We want you to apply your knowledge to something you care about. The blacklist encourages you to choose a topic you're genuinely interested in—whether that's a hobby, a campus issue, a side project, or just something you're curious about. This makes the work more engaging and helps you see how database concepts apply beyond textbook examples.

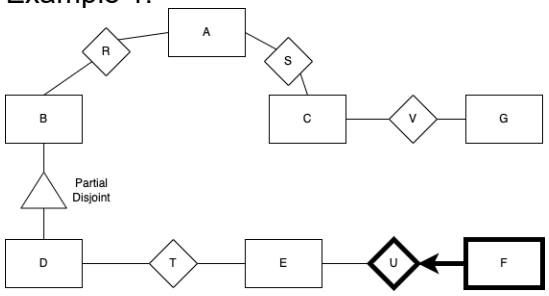
You don't need to reinvent the wheel—just pick a wheel you're actually interested in designing.

ER Diagram Requirements

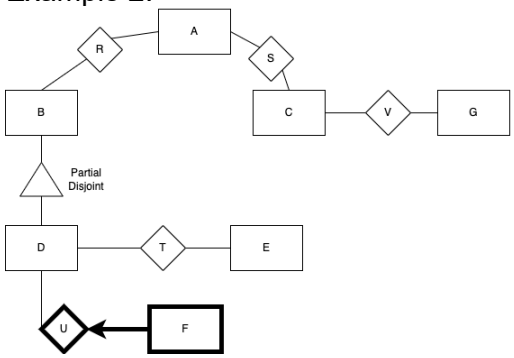
We expect each project to have:

- At least 7 different entities and 7 relationships, not counting weak entities and ISAs
    - All entities must have their keys clearly identified.
    - Aggregations are not counted. You can have them in your diagram but the aggregation itself will not count towards an entity or relationship.
    - What does “not counting weak entities and ISAs” mean? It means we only count owner/parent entity sets. For example, consider the following two examples (note that the examples are meant to illustrate how we count entity sets and do not include all the requirements we expect for the milestone 1 ER diagram).

Example 1:



Example 2:



Both example 1 and example 2 would be considered as five entity sets (specifically, we would count entity sets A, B, C, E, and G). Relationship set U does not count as one of the seven required relationships.

- At least one meaningful ISA relationship
    - An ISA relationship only counts as one entity. E.g., a parent entity with six children counts as one entity, not seven.
    - You must explicitly label your ISA relationship constraints (e.g., total + overlapping).
    - The child entities in an ISA must have their own unique and reasonable set of attributes and/or meaningfully participate in relationships that the other child entities do not.
- For each relationship, identify the cardinality constraint and other constraints, such as participation constraints. We encourage you to provide a mix of constraints (some 1-to-many, many-to-many, 1-to-1).
- All entities must have their keys clearly identified using the notation discussed in class
- All entities must have three meaningful non-primary key attributes. This will make milestone 2 a smoother experience.
- At least one meaningful (non-trivial) weak entity; but, if this is not suitable for your application, then replace it with one more meaningful ISA relationship.

Project Platforms

You may use any legal platform that you like, as long as:

- The final application uses a relational DBMS to manage the data.
- You don't use an automated system or tool (e.g., Github Copilot) to generate the code and SQL (we want you to write these on your own).
- You do not use libraries which will map relations to objects (sometimes referred to as ORM libraries).
- You do not use a GUI generating tool.
- All other requirements are met. Ask, if in doubt.
- You must use version control for your code. CPSC 304 will provision repositories for each team to use and you must periodically commit your code to this repository.

The department has an Oracle server available for your use (see here for more details). The course staff will be able to support PHP and JavaScript projects that use the department RDMBS. We will not be able to support any other combination of technology. However, some legacy instructions exist for Java/JDBC, which you are welcome to use, though the course staff may not be able to provide support with them.

If you choose to use something outside of the department provided infrastructure (e.g., a MySQL server on your own machine, or Java/JDBC), you may be asked to submit additional information. Furthermore, problems with non-supported platforms will NOT be accepted as an excuse for late project submission.

Deliverables

All the following items must be put together into a single PDF file.

1. A completed cover page (template on Canvas)

2. A brief project description answering these questions:
    a. What is the domain of the application? Describe it.

        The domain of an application refers to the area of knowledge your application resides in. For example, if I am making an application for a hospital, the domain would be something like healthcare/patient management/logistics (it would depend on what the application is trying to do).

        Ensure that all members of the team are familiar with the domain and data that will be modelled. If you do not understand the domain/data, you will have an exceptionally hard time meaningfully contributing to the project which will lead to **lots** of project issues down the road.

   b. What aspects of the domain are modeled by the database? In answering this question, you will want to talk about what your project is trying to address and how it fits within the domain. It is likely that in the process of answering these questions you will bring up examples of a real-life situation that the application could be applied to.

3. An ER diagram for the database that your application will use. It is OK to hand-draw it but if it is illegible or messy or confusing, marks will be taken off. You can use software to draw your diagram (e.g., draw.io, GoogleDraw, Microsoft Visio, Powerpoint, Gliffy, etc.) The result should be a legible PDF or PNG document. Note that your ER diagram must use the conventions from the textbook and the lectures. For example, do not use crow's feet notation or notation from other textbooks).
   a. Please limit your diagram to a letter size page (8.5 x 11 inches). If you require additional space, talk to your project mentor beforehand as this might mean that your project is a bit more complicated than what we expect.

4. Your E/R diagram should adhere to the expectations listed above.

5. Other comments, as appropriate, to explain your project.
6. An explicit acknowledgment about your use of AI tools in this assignment. Specifically, we are looking for a clear yes/no about whether you have used one or more AI tools. If yes, we want to know which tool(s) you have used and we want a PDF of the full conversation.

Check the Milestone 1 assignment on Canvas for the rubric. Refer to the syllabus for information on late submission/penalty rules.

FAQ

1. **Do we start coding from an empty repo? Where do we start coding once we are ready?**
   You can reuse the code provided in the sample project; in fact, this is recommended as opposed to starting from scratch. You will need to use the provisioned repository given to you by the CPSC 304 course staff.

   WARNING: Do not start coding until you get your feedback for milestone 2 as feedback from your TA may cause substantial changes.

2. **I want to ask my project mentor a question but I don't know who they are.**
   Project mentors will be assigned shortly after milestone 0 has concluded. However, as there are a lot of logistical issues behind the scenes, it will take a few days for those things to be settled. Once the project mentor assignment has been finalized, it will be released via Canvas. Keep an eye on Piazza for more information.

3. **Once we are finished with our milestone, does everyone in the group need to submit a copy or only one person?**
   Once you have been put into a group on Canvas (see the FAQ section in milestone 0), you can submit your work. Only one person needs to submit the deliverables; **always double check your submission to ensure that you have submitted the right documents and that the submission has gone through**. Reasons such as "I didn't realize I submitted the wrong file" are not considered as a valid reason for regrades or for waiving a late penalty.

   In some terms, there are milestones that need to be submitted individually so always double check. Milestones that need to be done individually will explicitly say so in the description document.

4. **Does our ER diagram need to be in colour like the examples on the lecture slides?**
   No. The lecture slides use colour in the ER diagrams to emphasize the difference between the various parts of the diagram. You do not need to follow that example. Having an ER diagram in grayscale is perfectly acceptable.

   Be sure to follow the notation used in class. We will not consider other types of notation as valid.